Topic 1[java Programming Basics]

Problem Statement: Write java programs to do the following

i) Display your name and address in two lines
ii) Take two integers as inputs and prints the bigger one
iii) Input a series of 10 numbers into an array ax[] and finds the largest, smallest and
average of these numbers.

Code	Output

<pre>import java.util.Scanner;
 	
public class Java_Programming_Basic
{
   public static void main(String kargs[])
   {
      Scanner scan=new Scanner(System.in);
 	
      // i)
      System.out.println("MD. ASAD-AL-ADIL");
      System.out.println("Rajshahi");
 	
      // ii)
      int a=scan.nextInt(),b=scan.nextInt();
      System.out.println("Max Number: "+(a>b?a:b));
 	
      // iii)
      int ax[]=new int[10];
      int sum=0;int mx=-100,mn=10000000;
      for(int i=0;i<10;i++)
      {
         System.out.print("Enter "+(i+1)+" number: ");
         ax[i]=scan.nextInt();
      }
       for(int i=0;i<10;i++)
      {
         sum=sum+ax[i];
         mx=mx>a?mx:ax[i];
         mn=mn<a?mn:ax[i];
      }
      System.out.println("The sum of 10 values is: "+sum);
      System.out.println("The MAX of 10 Values is: "+mx);
      System.out.println("The MAX of 10 Values is: "+mn);
      System.out.println(("The Average of 10 Values is: "+
         (float)sum/10)); // uses of type casting.......
 	
      scan.close();
   }
}</pre>
	i)
	MD. ASAD-AL-ADIL
	Rajshahi
	
	ii)
	10 15
	Max Number: 15
	
	iii)
	Enter 1 number: 1
	Enter 2 number: 2
	Enter 3 number: 3
	Enter 4 number: 4
	Enter 5 number: 5
	Enter 6 number: 6
	Enter 7 number: 7
	Enter 8 number: 8
	Enter 9 number: 9
	Enter 10 number: 10
	The sum of 10 values is: 55
	The MAX of 10 Values is: 10
	The MAX of 10 Values is: 1
	The Average of 10 Values is: 5.5
	Press any key to continue . . .

Topic 2[java Programming Access Modifier]

Problem Statement: Verify the following access modifier using the appropriate coding

Access Modifier	within class	within package	outside package by subclass only	outside package
Private	Y	N	N	N
Default	Y	Y	N	N
Protected	Y	Y	Y	N
Public	Y	Y	Y	Y

<table>
<tr><th>Code</th><th>Output</th></tr>
<tr><td>

```java
package school;
// within package
public class Teacher
{
    Student info=new Student();
    public Teacher()
    {

    }
    void show()
    {
        System.out.println(info.name);
        System.out.println(info.age);
        System.out.println(info.roll);
        System.out.println(info.result);
    }
}
```

</td><td>

We can see that there is indicating wrong sign on accessing private member, so it is not possible to do that but others can be accessible. Default, protected and public member can be accessible.
So-→N Y Y Y

</td></tr>
<tr><td>

```java
package school;
// within class
public class Student
{
    private String name="Tom";
    int age=17;
    public int roll=10;
    protected float result=4.95f;
    public Student()
    {

    }
    void show()
    {
        System.out.println(name);
        System.out.println(age);
        System.out.println(roll);
        System.out.println(result);
    }
}
```

</td><td>

All are accessible
So-→ Y Y Y Y..

</td></tr>
</table>

```java
package office;
import school.Student;

public class member extends Student
{
    member()
    {

    }
    void call()
    {
        System.out.println(name);
        System.out.println(age);
        System.out.println(roll);
        System.out.println(result);
    }
}
```

// Outside package of subclass only………

Here we can see that only protected and public member can accessible.
So-→ N N Y Y

```java
// Outside of package....
import school.Student;
public class access_modifier
{
    Student info=new Student();
    access_modifier()
    {

    }
    void call()
    {
        System.out.println(info.name);
        System.out.println(info.age);
        System.out.println(info.result);
        System.out.println(info.roll);
    }
}
```

Here we can see that only public method can be accessible.
So-→ N N N Y

Topic 3[java Programming private method]

Problem Statement: Verify that private methods can be used only inside the class but it can not be used from outside the class.

Inside class

```java
class Student
{
    public int roll=10;
    Student()
    {

    }
    private void show()
    {
        System.out.println("Roll is: "+10);
    }
    public void call()
    {
        show();
    }
}
```

Outside of class

```java
public class private_method
{
    Run | Debug
    public static void main(String Adil[])
    {
        Student info=new Student();
        info.show();
        info.call();
    }
}
```

From this table it is cleared that we can use private method inside of that class and giving no error but from outside of class we can call only public method creating as object and private method is not callable and showing error sign for this.

Topic 4[java Programming static method/member]

Problem Statement: Verify that static methods can access static method/member but can not access to non-static method/member.

<table>
<tr><th>Code</th><th>Output</th></tr>
<tr><td>

```
public class static_method
{
    public static void  print()
    {
        System.out.println(x:"Hello java");
    }
    public int sum()
    {
        int a=10,b=20;
        return a+b;
    }
    public static void call()
    {
        print();
        int x=sum();
    }
}
```

</td><td># Here we can see that static method call wants to access print method and sum method. But call() is a static method and it can access only static method print(). On the other hand sum() is a non-static method, so it cannot callable from static method.</td></tr>
</table>

Topic 5[java Programming string class]

Problem Statement: Verify the following methods of string class:

charAt() compareTo() concat() equals()	isEmpty() length() replace() substring()	toLowerCase() toUpperCase() toString() trim()

<table>
<tr><th>Code</th><th>Output</th></tr>
<tr><td>

```
public class string_class
{
   public static void main(String Adil[])
   {
       String s1="Java Programming ",s2="is good";
       String s3="";
       Integer x=1234567;
       System.out.println(s1.charAt(0));
       System.out.println(s1.compareTo(s2));
       s3=s1.concat(s2);
       System.out.println(s3);
       System.out.println(s1.equals(s2));
       System.out.println(s2.isEmpty());
       System.out.println(s1.length());
       System.out.println(s3.replace("Java","JAVA"));
       System.out.println(s3.substring(5, 16));
       System.out.println(s1.toLowerCase());
       System.out.println(s2.toUpperCase());
       s3=x.toString();
       System.out.println(s3.charAt(2));
       String s4="    Hello,Java      ";
       String s5=s4.trim();
```

</td><td>

```
J
-31
Java Programming is good
false
false
17
JAVA Programming is good
Programming
java programming
IS GOOD
3
Hello,Java
Press any key to continue . . .
```

</td></tr>
</table>

System.out.println(s5); } }	

Topic 6[java Programming anonymous object]

Problem Statement: Write a program to do the following using array

i. write a method **InitiaArray()** that returns the initial array as anonymous object. ii. write a method **SortArray()** that returns the sorted array as anonymous object. iii. write a method **PrintArray()** that prints the array

Code	Output

<pre>import java.util.Arrays;
 	
public class anonymous_object
{
   public static void main(String Adil[])
   {
      int A[]=InitialArray();
      for(int i:A)
      {
         System.out.print(i+" ");
      }
      System.out.println();
      int B[]=SortArray();
      for(int i:B)
      {
         System.out.print(i+" ");
      }
      System.out.println();
   }
   public static int[] InitialArray()
   {
      return new int[]{2,4,6,1,3,5};
   }
   public static int[] SortArray()
   {
      int A[]=InitialArray();
      Arrays.sort(A);
      return A;
   }
}</pre>
	2 4 6 1 3 5
	
	1 2 3 4 5 6
	
	Press any key to continue . . .

Topic 7[**java Programming toString() & (…)**]

Problem Statement: Write a program to do the following using array

i. display the array elements using **toString()** method ii. write a method **FindBig()** that returns the largest of first four numbers of the array using (...) variable length parameter.

Code	Output

public class tostring
{
public static void main(String Adil[])
{
int ax[]={1,2,3,4,5,6,7};
toString(ax); //i)
System.out.println(FindBig(ax[0],ax[1],ax[2],ax[3]));
// ii)
}
public static void toString(int []ax)
{
for(int i:ax)
{
System.out.println(i);
}
System.out.println();
}
public static int FindBig(int...nums)
{
int x=-10;
for(int i=0;i<4;i++)
{
x=nums[i]>x?nums[i]:x;
}
return x;
}
}
	1
	2
	3
	4
	5
	6
	7
	
	4
	Press any key to continue . . .

Topic 8[**java Programming ArrayList class**]

Problem Statement: Write a program to create an array **ax** using AarryList class of integers. Then for the array **ax**.

i.add 4 elements 10, 50, 70 and 30 using add() method ii. insert 60 at index 1 iii. print element at index 2 iv. delete element of index 3 v. delete element 10 vi. sort and prints the elements using for-each loop vii. delete all the elements of the array

Code	Output
import java.util.ArrayList; public class arraylist { public static void main(String Adil[])	10 50 70 30 10 60 50 70 30

```
  {
     ArrayList<Integer>ax=new ArrayList<>();
     ax.add(10);
     ax.add(50);
     ax.add(70);
     ax.add(30);
     Display(ax); //i)
     ax.add(1,60);
     Display(ax); //ii)
     System.out.println(ax.get(2)); //iii)
     ax.remove(3);
     Display(ax); // iv)
     ax.remove(ax.indexOf(10));
     Display(ax); // v)
     ax.sort(null);
     Display(ax); // vi)
     ax.clear();
     System.out.println(ax.size());

  }
  public static void Display(ArrayList<Integer>ax)
  {
     for(int i:ax)
     {
        System.out.print(i+" ");
     }
     System.out.println();
  }
}
```

```
50

10 60 50 30

60 50 30

30 50 60

0
Press any key to continue . . .
```